

Causal Inference in Economics and Marketing

Section 10: Mediation and Mechanisms

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1 From effects to mechanisms

1.1 The question behind the average effect

Sections 02 to 08 asked whether a treatment works and for whom. This section asks how. Two published experiments show why the question is different.

Wheeler, Garlick, Johnson, Shaw, and Gargano (2022) randomized a LinkedIn training module inside a job-readiness program in South Africa. Employment at the end of the program rose by 7 percentage points. The obvious story is that the training raised LinkedIn use and LinkedIn use produced jobs. The experiment identifies the first step: treatment raised account ownership by 32 percentage points. It does not identify the second step. Nobody randomized LinkedIn use. The paper says so and then reports two “share explained” statistics under stated assumptions.

Cattan, Salvanes, and Tominey (2025) study exposure to elite peers in Norwegian high schools. Exposure raises applications to elite programs. It also lowers grades, and grades matter for admission. The total effect is the sum of a positive direct channel and a negative channel through grades. To separate them the authors instrument grades with a lottery that assigns students to blind written mathematics exams.

Both papers need a language for “part of the effect runs through M ”, and both need assumptions beyond the experiment. This section provides the language, the assumptions, and the estimators.

1.2 Why “control for the mediator” is not an answer

The first instinct is to add the mediator to the regression and read the coefficient on D as the direct effect. Figure 1 shows what that does. The design is linear: D is randomized, M depends on D , and Y depends on D and M . An unmeasured variable U affects both M and Y with a strength we vary from zero upward. The true direct effect is 1 and the total effect is $1 + 1.2 \times 0.9 = 2.08$.

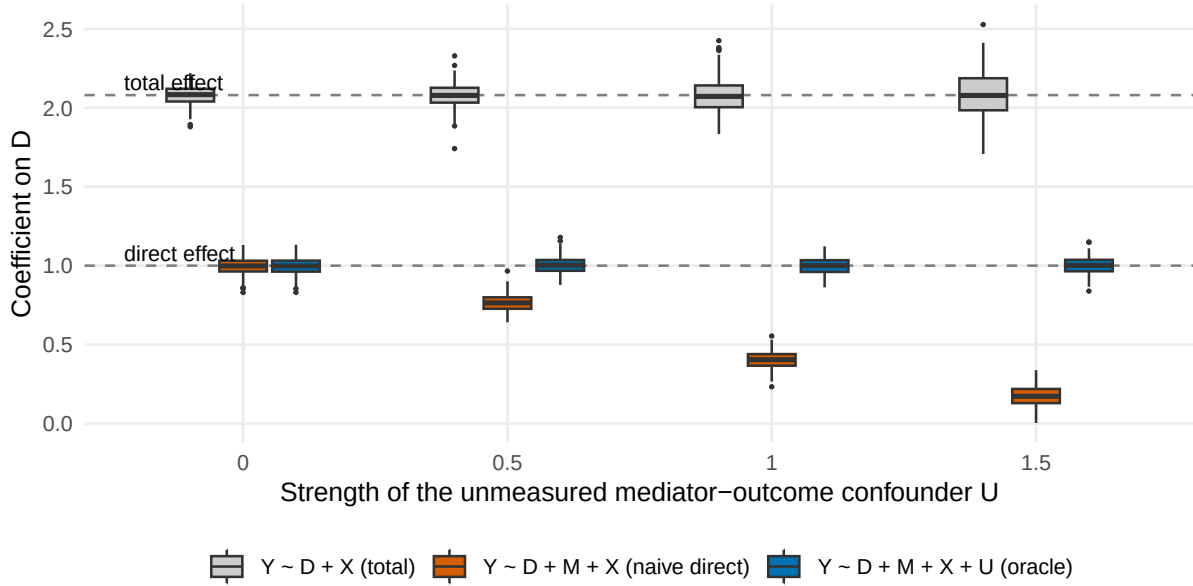


Figure 1: The coefficient on D across repeated samples ($n = 2000$) when the mediator is added to the regression. D is randomized, so the total-effect regression is right at every strength of the unmeasured mediator-outcome confounder U . The regression that adds M is right only when U is absent. The oracle regression that also adds U is always right, but U is not observed.

With U absent the regression with M gives 1.00, the direct effect. At strength one it gives 0.40, and the error grows with the strength of U . Randomization of D does not help. D is exogenous, M is not, and once M is in the regression the coefficient on D is a partial coefficient in a system where M carries U into the equation. Section 2 gives the graphical reason: conditioning on M opens the path $D \rightarrow M \leftarrow U \rightarrow Y$.

The lesson is not “never condition on M ”. It is that the coefficient on D with M held fixed is a different estimand, and it is identified only under assumptions about M that no experiment on D delivers.

1.3 Three questions and three estimands

Write $M(d)$ for the mediator under treatment d and $Y(d, m)$ for the outcome under treatment d and mediator level m . Three questions map to three estimands.

Question	Estimand	Name
What does the treatment do?	$E[Y(1, M(1)) - Y(0, M(0))]$	total effect
What would it do if the mediator were held at its untreated value?	$E[Y(1, M(0)) - Y(0, M(0))]$	natural direct effect
What does the treatment do through the mediator alone?	$E[Y(1, M(1)) - Y(1, M(0))]$	natural indirect effect

The first estimand needs an experiment on D . The other two contain $Y(1, M(0))$: the outcome of a treated unit whose mediator takes the value it would have taken without treatment. No unit ever displays this. The

literature calls such quantities cross-world. Identifying them requires either an assumption that makes the mediator as good as randomized given observables, or a design that moves the mediator on its own.

1.4 Roadmap

Section 2 gives the graphical vocabulary. Section 3 defines the estimands with care, including the controlled direct effect and the interaction case. Section 4 states sequential ignorability, proves the mediation formula, and lists the designs that identify mediation without it. Sections 5 to 7 estimate: the linear system, regression with interaction and simulation, and doubly robust machine learning. Section 8 covers the regression decompositions applied papers use. Section 9 treats mediators with their own exogenous variation: instruments and the front door. Section 10 covers several mediators and mediators built from text. Sections 11 and 12 close with sensitivity, design, and a checklist. Proofs are in the appendix.

2 Graphs for mechanisms

2.1 Chains, forks, colliders

A directed acyclic graph (DAG) has one node per variable and one arrow per direct causal effect. Three patterns explain every rule we need (Chernozhukov et al. 2026, ch. 7; Lattimore, MS&E 228 lecture 9).

- **Chain** $D \rightarrow M \rightarrow Y$. D and Y are dependent. Conditioning on M blocks the path.
- **Fork** $D \leftarrow X \rightarrow Y$. D and Y are dependent through the common cause. Conditioning on X blocks the path.
- **Collider** $D \rightarrow M \leftarrow U$. D and U are independent. Conditioning on M makes them dependent.

A path is blocked if it contains a non-collider that is conditioned on, or a collider that is not conditioned on (and none of whose descendants is conditioned on). Two variables are d-separated given a set S when every path between them is blocked given S . In a DAG compatible with the data, d-separation implies conditional independence.

The collider rule is the one that bites in mediation. In Figure 2, right panel, D and U are independent. Among units with the same M , a high D must go with a low U or M would not be the same. Conditioning on M creates a negative association between D and U , and U drives Y . That is the mechanism behind Figure 1.

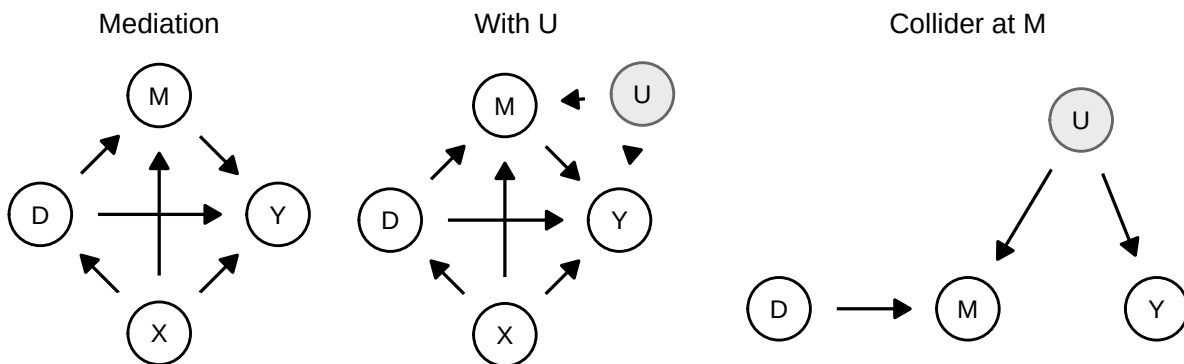


Figure 2: Left: the mediation graph with pre-treatment covariates X . Centre: the same graph with an unmeasured mediator-outcome confounder U (shaded). Right: the collider structure that conditioning on M activates: D and U are independent, but not among units with the same M .

2.2 The mediation graph and its confounders

The centre panel of Figure 2 is the object of the section. Four kinds of confounding can sit on it.

1. D - Y confounding: a common cause of the treatment and the outcome. Randomization removes it.
2. D - M confounding: a common cause of the treatment and the mediator. Randomization removes it.
3. M - Y confounding: a common cause of the mediator and the outcome, the U of the figure. Randomization of D does nothing about it.
4. M - Y confounding by a variable that the treatment itself affects, L in Figure 3. This is the hardest case.

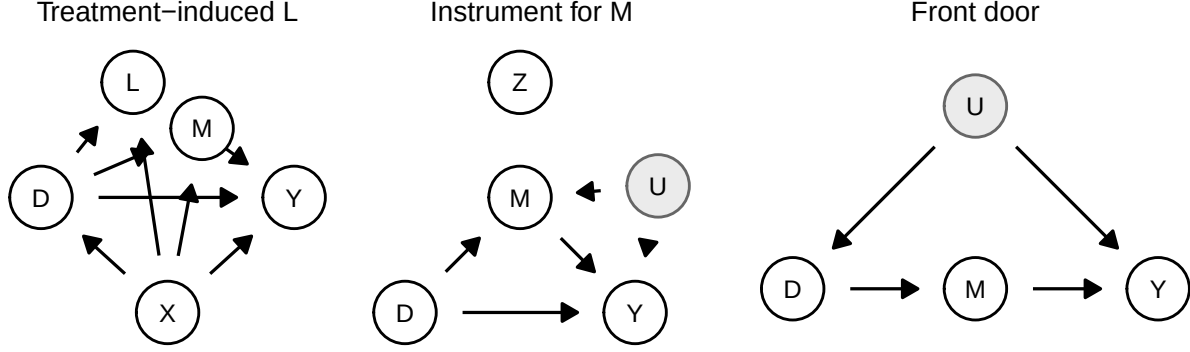


Figure 3: Left: a treatment-induced confounder L of the mediator and the outcome. Conditioning on L is needed for the M - Y relation and forbidden for the D - Y relation. Centre: an instrument Z for the mediator. Right: the front-door structure, where the D - Y confounder U is unobserved but M is unconfounded and carries the whole effect.

2.3 Good and bad controls

Lecture 10 of MS&E 228 organizes the adjustment question around one rule: a set S identifies the total effect of D on Y if it blocks every back-door path (paths that start with an arrow into D) and contains no descendant of D . Pre-treatment covariates that cause both D and Y are good controls. Post-treatment variables are bad controls for the total effect, for two reasons.

- A mediator blocks part of the effect. The coefficient on D then measures something between the total and the direct effect.
- A collider opens a path. Conditioning on M in Figure 2 opens $D \rightarrow M \leftarrow U \rightarrow Y$.

`dagitty` computes adjustment sets from a graph. For the total effect of D in the graph with U , the only valid set is $\{X\}$; for the effect of M on Y it needs U , which is unobserved.

```
g <- dag_mediation_u()
dagitty::adjustmentSets(g, exposure = "D", outcome = "Y")
#> { X }
dagitty::adjustmentSets(g, exposure = "M", outcome = "Y")
#> { D, U, X }
```

The wage-gap example of Chernozhukov et al. (2026, ch. 6) is the same question in different clothes. The gender coefficient in a wage regression falls when occupation and experience are added. Occupation and experience are post-treatment for gender. The drop is not a direct effect. It is the part of the gap that the added variables absorb, and it is a mechanism claim only under the assumptions of Section 4.

2.4 Structural equations and potential outcomes

A nonparametric structural equation model writes each variable as a function of its parents and an error:

$$M = f_M(D, X, e_M), \quad Y = f_Y(D, M, X, e_Y).$$

Potential outcomes are obtained by substitution: $M(d) = f_M(d, X, e_M)$, $Y(d, m) = f_Y(d, m, X, e_Y)$, and the composition $Y(d, M(d')) = f_Y(d, f_M(d', X, e_M), X, e_Y)$. Consistency links them to data: $M = M(D)$ and $Y = Y(D, M)$. The composition with $d \neq d'$ is the cross-world quantity. It is well defined in the model, but the data only show $Y(d, M(d))$.

3 Estimands

3.1 Total effect and controlled direct effect

The total effect is $TE = E[Y(1, M(1)) - Y(0, M(0))] = E[Y(1) - Y(0)]$. The controlled direct effect at mediator level m is

$$CDE(m) = E[Y(1, m) - Y(0, m)].$$

It answers a policy question: if we could fix the mediator at m for everyone, what would the treatment do? It depends on m whenever the treatment and the mediator interact, and it does not decompose the total effect. Its identification only needs the two intervention-world assumptions of Section 4, which is why it survives treatment-induced confounders.

3.2 Natural direct and indirect effects

The natural direct effect holds each unit's mediator at its own untreated value; the natural indirect effect moves the mediator while holding the treatment (Robins and Greenland 1992; Pearl 2001):

$$NDE = E[Y(1, M(0)) - Y(0, M(0))], \quad NIE = E[Y(1, M(1)) - Y(1, M(0))].$$

Adding and subtracting $E[Y(1, M(0))]$ gives $TE = NDE + NIE$. Two other quantities come from the opposite reference levels:

$$NDE^{tot} = E[Y(1, M(1)) - Y(0, M(1))], \quad NIE^{pure} = E[Y(0, M(1)) - Y(0, M(0))],$$

and $TE = NDE^{tot} + NIE^{pure}$ as well. VanderWeele (2015) calls NDE the pure direct effect and NIE the total indirect effect. When the treatment and the mediator do not interact the two decompositions coincide. When they do, the difference $NDE^{tot} - NDE$ is the mediated interaction, the part of the effect that needs both the treatment and the mediator change. Reporting both decompositions is cheap and avoids an arbitrary choice.

The proportion mediated NIE/TE is popular and fragile. It is a ratio, so its standard error is large when the total effect is small, and it is outside $[0, 1]$ whenever the direct and indirect effects have opposite signs, as in Cattani et al. (2025).

3.3 Path coefficients in linear systems

In the linear system

$$M = \alpha_0 + \alpha_1 D + X' \alpha_X + e_M, \quad Y = \beta_0 + \beta_1 D + \beta_2 M + X' \beta_X + e_Y,$$

substitution gives $Y(d, M(d')) = \beta_0 + \beta_1 d + \beta_2(\alpha_0 + \alpha_1 d' + X' \alpha_X + e_M) + X' \beta_X + e_Y$, so

$$NDE = \beta_1, \quad NIE = \alpha_1 \beta_2, \quad TE = \beta_1 + \alpha_1 \beta_2, \quad CDE(m) = \beta_1 \text{ for every } m.$$

Wright's rules for path analysis say the same: effects along a path multiply, effects across paths add. Figure 4 labels the edges.

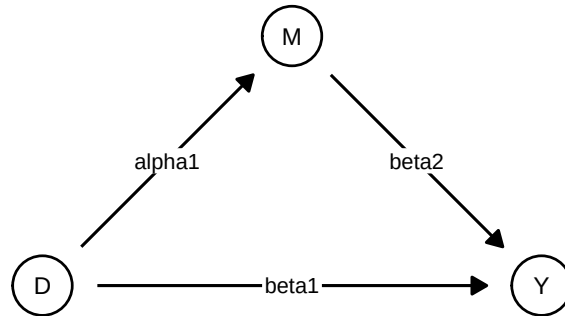


Figure 4: Path coefficients in the linear mediation model. The indirect effect is the product along the path through M , the direct effect is the coefficient on the direct edge, and the total effect is their sum.

3.4 Wheeler’s shares in this language

Wheeler et al. (2022, Appendix C) report two shares of the employment effect explained by LinkedIn use. Their system is the linear one above with β the coefficient on treatment in $Y \sim D$, γ the treatment effect on the mediator, α the mediator coefficient in $Y \sim D + M$, and δ the mediator coefficient in the control group only:

$$S_1 = \frac{\alpha\gamma}{\beta}, \quad S_2 = \frac{\delta\gamma}{\beta}.$$

S_1 is the proportion mediated of the linear model: $\alpha\gamma$ is the NIE and β the total effect. S_2 replaces the mediator effect by its control-group value. Under the linear model without interaction $\alpha = \delta$ in population, and the two shares differ only by sampling error. When the mediator effect differs by treatment status, δ is the effect at the untreated level, so S_2 corresponds to the pure indirect effect over the total. Both shares carry the same identification burden: the mediator coefficient is causal only if the mediator is unconfounded given D and the controls.

4 Identification

4.1 Sequential ignorability

Imai, Keele, and Yamamoto (2010) state the assumption in two parts. For all d, d', m and x in the support,

$$(SI-1) \quad \{Y(d', m), M(d)\} \perp D \mid X = x, \quad (SI-2) \quad Y(d', m) \perp M(d) \mid D = d, X = x,$$

with $0 < P(D = d \mid X = x)$ and $0 < p(M(d) = m \mid D = d, X = x)$. SI-1 says the treatment is ignorable given pre-treatment covariates. An experiment delivers it. SI-2 says that among units with the same treatment and covariates, the mediator is ignorable with respect to the potential outcomes. No experiment on D delivers it. It rules out the U of Figure 2 and the L of Figure 3: X must be pre-treatment, so a treatment-induced confounder cannot be in it.

Pearl (2001) and VanderWeele (2015) state four conditions: no unmeasured D - Y , D - M , and M - Y confounding given X , and no M - Y confounder affected by D . SI-1 and SI-2 together imply them, and for the estimators of this section the two sets are interchangeable.

4.2 The mediation formula

Proposition 1. Under sequential ignorability, for every d, d' ,

$$E[Y(d, M(d'))] = \int \int E[Y \mid D = d, M = m, X = x] dF_{M \mid D=d', X=x}(m) dF_X(x).$$

Skeleton. Fix x . SI-1 lets us replace the distribution of $M(d')$ by the observed distribution of M among units with $D = d'$. SI-2 lets us replace $E[Y(d, m) \mid M(d') = m, D = d, X = x]$ by $E[Y(d, m) \mid D = d, X = x]$: the potential outcome does not depend on which mediator value the unit would have had. SI-1 and consistency then give $E[Y \mid D = d, M = m, X = x]$. Integrate over m and x . The formal proof is in Appendix 13.1.

Every estimator in Sections 5 to 7 is an estimator of this integral: plug-in regressions (Section 5), simulation of the integral (Section 6), and an influence-function estimator whose bias is second order in the nuisance errors (Section 7).

4.3 When it fails

Unmeasured M - Y confounding. SI-2 fails, and Figure 1 shows the consequence. Nothing in the data reveals the failure, so the response is sensitivity analysis: how strong must U be to explain the estimated indirect effect (Section 6.3).

Treatment-induced confounding. When L in Figure 3 exists, the natural effects are not identified without further restrictions, even with L observed. Avin, Shpitser, and Pearl (2005) call L a recanting witness:

identification of the path $D \rightarrow M \rightarrow Y$ needs L held fixed on the $D \rightarrow L \rightarrow M$ part and free on the $D \rightarrow L \rightarrow Y$ part, and no adjustment does both. The controlled direct effect survives, because it only needs the intervention-world assumptions $Y(d, m) \perp D \mid X$ and $Y(d, m) \perp M \mid D, L, X$. Section 7.4 estimates it.

Interaction is not a failure. The mediation formula holds with any interaction between D and M . What changes is that the product of coefficients is no longer the right plug-in.

4.4 Designs that identify mediation

Imai, Tingley, and Yamamoto (2013) list designs that replace SI-2 with experimental variation in the mediator.

- **Parallel design.** One arm randomizes D only; another randomizes D and sets M . The second arm identifies $E[Y(d, m)]$; together with the first it bounds the natural effects, and identifies them if the treatment does not interact with the mediator at the unit level.
- **Crossover design.** Each unit is observed under $(d, M(d))$ and then under $(d, M(1 - d))$ with the mediator set to its earlier value in the other arm. It needs no carryover.
- **Encouragement design.** The mediator is not set but encouraged, with an instrument Z . This identifies effects among mediator compliers, and the full decomposition under a homogeneity assumption. Cattani et al. (2025) use the exam lottery as the encouragement (Section 9.1).
- **Front-door structure.** When the treatment affects the outcome only through the mediator and the mediator is unconfounded given D , the effect is identified even with unmeasured D - Y confounding (Section 9.2).

The first two are rare outside the laboratory. The last two are the applied economist's tools.

5 Estimation I: the linear system

5.1 Product of coefficients and the delta method

The running example is the linear design of `sim_mediation()`. The mediator is $M = 0.5 + 1.2D + 0.6x_1 - 0.4x_2 + e_M$ and the outcome is $Y = 1 + D + 0.9M + 0.5x_1 + 0.3x_2 + e_Y$; the treatment probability depends on x_1 and x_2 . The truth is $NDE = 1$, $NIE = 1.08$, $TE = 2.08$.

```
dat <- sim_mediation(3000, dgp = "linear", seed = 1)
fit_lin <- mediate_reg(dat, "y", "d", "m", x = c("x1", "x2"), method = "delta")
fit_lin$effects[fit_lin$effects$term %in% c("total", "nde", "nie"), ]
#>      term estimate std.error conf.low conf.high
#> 1 total 2.088636 0.04948124 1.9916550 2.185618
#> 2 nde 1.036152 0.04295957 0.9519532 1.120352
#> 3 nie 1.052484 0.03882209 0.9763941 1.128574
```

The indirect effect 1.052 is exactly the product of the treatment coefficient in the mediator regression (1.204) and the mediator coefficient in the outcome regression (0.874). Its standard error is Sobel's (1982) delta-method formula

$$\widehat{se}(\hat{\alpha}_1\hat{\beta}_2) = \sqrt{\hat{\alpha}_1^2 \widehat{var}(\hat{\beta}_2) + \hat{\beta}_2^2 \widehat{var}(\hat{\alpha}_1)},$$

which treats the two regressions as independent (Appendix 13.3). The simulation method of Section 6 and the bootstrap give the same answer here to two decimals, and they also work when the models are nonlinear.

```
fit_sim <- mediate_reg(dat, "y", "d", "m", x = c("x1", "x2"), method = "simulation", seed = 1)
fit_boot <- mediate_reg(dat, "y", "d", "m", x = c("x1", "x2"), method = "bootstrap", n_boot = 199, seed = 1)
tibble::tibble(method = c("delta", "simulation", "bootstrap"),
  nie = c(fit_lin$effects$std.error[3], fit_sim$effects$std.error[3], fit_boot$effects$std.error[3]))
#> # A tibble: 3 x 2
#>   method      nie
#>   <chr>      <dbl>
#> 1 delta      0.0388
#> 2 simulation 0.0395
#> 3 bootstrap 0.0423
```


5.2 Wheeler’s shares on the LinkedIn data

Appendix Table C1 of Wheeler et al. (2022) reports, for the LinkedIn account mediator, $\beta = 0.070$, $\gamma = 0.321$, $\alpha = 0.158$, $\delta = 0.146$, $S_1 = 0.729$ (0.306) and $S_2 = 0.672$ (0.266), with stratification-block fixed effects and standard errors clustered by cohort. `mediate_reg()` reproduces the table from the public data: the shares are in `shares`, and their standard errors come from the delta method on the stacked influence functions of the four regressions, which carries the cross-equation covariance as Stata’s `suest` does.

```
wh <- as.data.frame(haven::read_dta(wheeler_path))
wh$strata <- factor(wh$strata)
wh_acc <- wh[complete.cases(wh[, c("employed1_all", "treatment", "li_account")]), ]
fit_wh <- mediate_reg(wh_acc, "employed1_all", "treatment", "li_account", x = "strata",
                     method = "delta", cluster = "studref")

fit_wh
#> Causal mediation by regression (linear outcome, linear mediator; inference: delta)
#>   n = 1626; treatment treatment; mediator: li_account
#>      term estimate std.error conf.low conf.high
#>      total  0.06986   0.02051  0.02966  0.11006
#>      nde    0.01897   0.02607 -0.03213  0.07006
#>      nie    0.05089   0.01164  0.02808  0.07371
#> nde_total  0.01897   0.02607 -0.03213  0.07006
#> nie_pure   0.05089   0.01164  0.02808  0.07371
#>      cde    0.01897   0.02607 -0.03213  0.07006
#> proportion mediated = 0.729 (0.306)
#> S1 (alpha*gamma/beta) = 0.729 (0.306)
#> S2 (delta*gamma/beta) = 0.672 (0.266)
```

Table 2: Appendix Table C1 of Wheeler et al. (2022) and its reproduction with `mediate_reg()`. Columns 2-3: LinkedIn account as the mediator; columns 4-5: the LinkedIn usage index. Standard errors clustered by cohort; block fixed effects in every regression.

	LinkedIn account		Usage index	
	Paper	causalmetrics	Paper	causalmetrics
Treatment effect on employment (beta)	0.070	0.070	0.083	0.083
Treatment effect on LinkedIn use (gamma)	0.321	0.321	0.954	0.954
Treatment effect on employment given LinkedIn use (beta tilde)	0.019	0.019	0.028	0.028
Employment-LinkedIn association given treatment (alpha)	0.158	0.158	0.058	0.058
Employment-LinkedIn association in the control group (delta)	0.146	0.146	0.059	0.059
S1 = alpha gamma / beta	0.729	0.729	0.668	0.668
(se)	(0.306)	(0.306)	(0.253)	(0.253)
S2 = delta gamma / beta	0.672	0.672	0.678	0.678
(se)	(0.266)	(0.266)	(0.259)	(0.259)
Sample size	1626	1626	1288	1288

The point estimates and the standard errors match the paper to three decimals. The reading is where the two papers diverge from the two functions: the shares say that, if LinkedIn use were as good as randomly assigned among trainees with the same block, about two thirds of the employment gain would run through it. The paper labels this a non-prespecified accounting under strong assumptions, and Section 6.3 quantifies how strong.

5.3 Where linearity breaks

The `interaction` design adds $0.8 DM$ to the outcome equation. The product of coefficients has no place for this term. Table 3 compares the linear plug-in with the regression that includes the interaction.

Without the interaction the total effect is still right, because $Y \sim D + X$ needs no mediator model, but the split is wrong: the plug-in NIE misses the truth by -0.51. With the interaction the two decompositions differ by 0.95, the mediated interaction, and the controlled direct effect at $m = 0$ is a different number from either natural direct effect.

Table 3: The interaction design: truth, the product-of-coefficients plug-in, and the regression with a treatment-mediator interaction (simulation standard errors).

estimand	truth	no interaction	with interaction	se
Total	3.456	3.444	3.444	0.065
NDE (pure)	1.416	1.910	1.425	0.052
NIE (total)	2.040	1.534	2.019	0.071
NDE (total)	2.376	1.910	2.378	0.052
NIE (pure)	1.080	1.534	1.065	0.045
CDE(0)	1.000	1.910	1.006	0.054

6 Estimation II: regression with interaction and simulation

6.1 Closed forms and the scale problem

With the outcome model $Y = \beta_0 + \beta_1 D + \beta_2 M + \beta_3 DM + X' \beta_X + e_Y$ and the linear mediator model, substitution into the mediation formula gives (VanderWeele 2015, ch. 2)

$$\begin{aligned} NDE &= \beta_1 + \beta_3 E[\alpha_0 + X' \alpha_X], & NIE &= (\beta_2 + \beta_3) \alpha_1, \\ NDE^{tot} &= \beta_1 + \beta_3 E[\alpha_0 + \alpha_1 + X' \alpha_X], & NIE^{pure} &= \beta_2 \alpha_1, & CDE(m) &= \beta_1 + \beta_3 m. \end{aligned}$$

`mediate_reg(interaction = TRUE)` computes them by g-computation rather than by these formulas, so that the same code handles logistic models.

Binary outcomes raise a scale question. A logistic outcome model has coefficients on the log-odds scale, and log-odds do not average: the population NDE on the risk-difference scale is not a function of the coefficients alone. VanderWeele and Vansteelandt (2010) give closed forms on the odds-ratio scale under a rare-outcome approximation. The package avoids the approximation by averaging predicted probabilities over the sample, which is the mediation formula on the probability scale. The `binary` design has a logistic mediator and a logistic outcome.

```
dat_bin <- sim_mediation(3000, dgp = "binary", seed = 3)
fit_bin <- mediate_reg(dat_bin, "y", "d", "m", x = c("x1", "x2"),
                      outcome = "logit", mediator = "logit", n_sim = 500, seed = 1)
cbind(fit_bin$effects[1:3, 1:3], truth = unlist(attr(dat_bin, "truth")[c("total", "nde", "nie"))])
#>   term estimate std.error truth
#> 1 total 0.2745198 0.018483662 0.25300000
#> 2 nde 0.1778293 0.018792395 0.15733333
#> 3 nie 0.0966905 0.008649908 0.09566667
```

6.2 The simulation algorithm as g-computation

Imai, Keele, and Tingley (2010) estimate the mediation formula in three steps.

1. Fit the mediator model and the outcome model; draw their coefficients from the normal approximation to their sampling distribution (a quasi-Bayesian step).
2. For each draw and each unit, compute the mediator's distribution under $d' = 0$ and $d' = 1$, then the expected outcome under each (d, d') pair, integrating over the mediator when the outcome model is nonlinear.
3. Average over units to get $E[Y(d, M(d'))]$ for the draw; the effects are differences; their spread across draws gives the intervals.

This is g-computation (Robins 1986) with parametric models and a simulation-based variance. The `mediation` package of Tingley et al. (2014) implements it; `mediate_reg()` does the same with the interaction and logit variants and reports both decompositions. Figure 5 shows the draws behind the intervals of the interaction design.

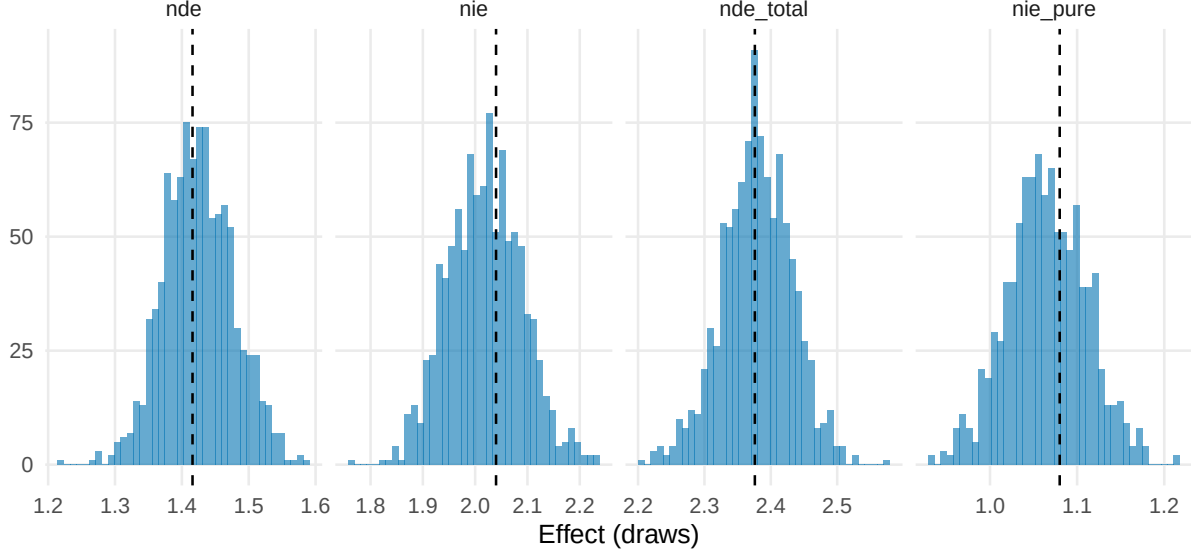


Figure 5: The quasi-Bayesian draws behind the intervals: 1000 draws of the coefficients produce 1000 values of each effect; the bars are the 2.5 and 97.5 percent quantiles. Dashed lines are the truth of the interaction design.

6.3 Sensitivity to an unmeasured mediator-outcome confounder

Imai, Keele, and Yamamoto (2010, Theorem 3) parameterize the violation of SI-2 in the linear model by the correlation ρ between e_M and e_Y . The indirect effect as a function of ρ is

$$NIE(\rho) = \beta_2 \frac{\sigma_1}{\sigma_2} \left(\tilde{\rho} - \rho \sqrt{\frac{1 - \tilde{\rho}^2}{1 - \rho^2}} \right),$$

where β_2 is here the treatment effect on the mediator, σ_1 and σ_2 are the residual standard deviations of $Y \sim D + X$ and $M \sim D + X$, and $\tilde{\rho}$ is the correlation of their residuals. At $\rho = 0$ the formula gives the product-of-coefficients estimate. The value ρ_0 where $NIE(\rho_0) = 0$ is the sensitivity parameter: an unmeasured confounder must induce an error correlation of at least ρ_0 in absolute value to erase the indirect effect.

The `correlated_errors` design generates $\rho = 0.5$. The estimate at $\rho = 0$ is biased upward; the curve passes through the truth at the true ρ .

```
dat_rho <- sim_mediation(4000, dgp = "correlated_errors", rho = 0.5, seed = 5)
fit_rho <- mediate_reg(dat_rho, "y", "d", "m", x = c("x1", "x2"), method = "delta")
sens <- mediate_sensitivity(fit_rho, rho = seq(-0.8, 0.8, by = 0.05))
c(nie_at_0 = sens$curve$nie[sens$curve$rho == 0], nie_at_true = sens$curve$nie[sens$curve$rho == 0.5],
  truth = attr(dat_rho, "truth")$nie, rho_zero = sens$rho_zero)
#>      nie_at_0 nie_at_true      truth      rho_zero
#>  1.7022595  1.0873310  1.0800000  0.8477644
```

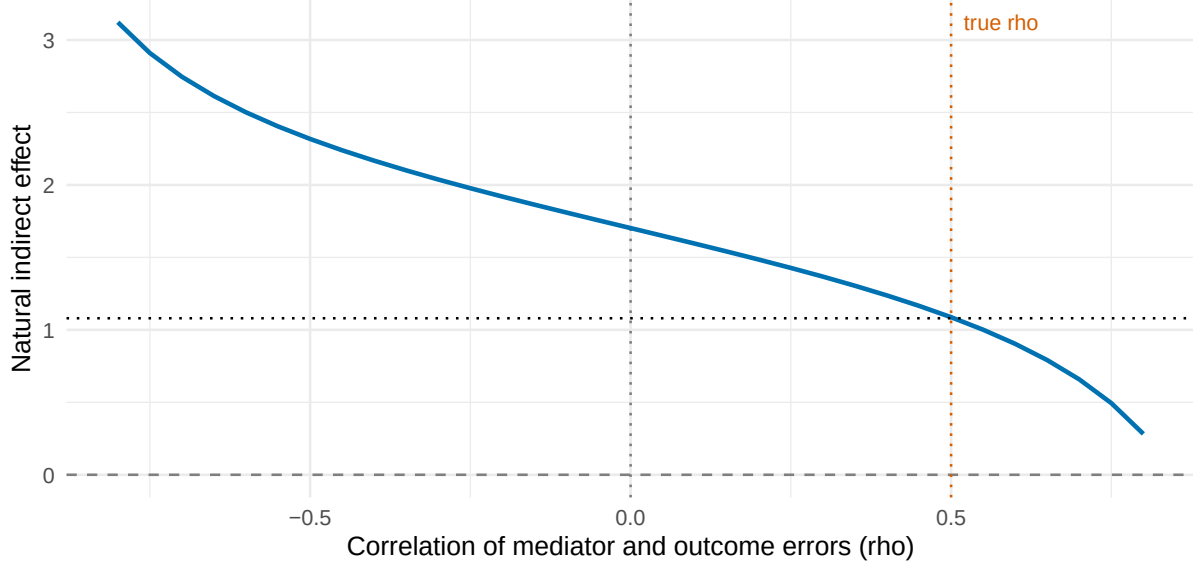


Figure 6: The indirect effect as a function of the error correlation ρ in the correlated-errors design. At $\rho = 0$ (sequential ignorability) the estimate is the product of coefficients; the curve crosses the truth (dotted) at the design's $\rho = 0.5$ and crosses zero at the value reported as rho zero.

```
sens_wh <- mediate_sensitivity(fit_wh, rho = seq(-0.5, 0.5, by = 0.05))
c(rho_zero = sens_wh$rho_zero, rho_tilde = sens_wh$rho_tilde)
#> rho_zero rho_tilde
#> 0.1633502 0.1633512
```

On the LinkedIn data the indirect effect through account ownership vanishes at an error correlation of 0.16. An unmeasured trait that raises both account opening and employment, with that much residual correlation, would explain the whole share. That is a modest amount for a labour-market outcome, which is the quantitative version of the paper's caution.

7 Estimation III: doubly robust mediation with machine learning

7.1 Why two nuisances are not enough

Section 04 estimated the average treatment effect with two nuisances, the propensity score and the outcome regression, and an influence function whose bias is the product of their errors. The mediation formula has a third layer. For $E[Y(1, M(0))]$ we need the outcome regression under treatment, $\mu_1(M, X) = E[Y \mid D = 1, M, X]$, averaged over the mediator distribution of the controls:

$$\nu_{1,0}(X) = E[\mu_1(M, X) \mid D = 0, X].$$

This nested regression is a regression of a fitted value. Tchetgen Tchetgen and Shpitser (2012) derive the efficient influence function for $E[Y(d, M(d'))]$, and Farbmacher, Huber, Laffers, Langen, and Spindler (2022) write it with propensity scores in place of mediator densities, so that the mediator can be continuous or multivariate:

$$\psi_{d,d'} = \frac{1\{D = d\} p_{d'}(M, X)}{p_d(M, X) p_{d'}(X)} (Y - \mu_d(M, X)) + \frac{1\{D = d'\}}{p_{d'}(X)} (\mu_d(M, X) - \nu_{d,d'}(X)) + \nu_{d,d'}(X),$$

with $p_d(X) = P(D = d \mid X)$ and $p_d(M, X) = P(D = d \mid M, X)$. The ratio $p_{d'}(M, X)/(p_d(M, X) p_{d'}(X))$ equals $f(M \mid d', X)/(f(M \mid d, X) p_{d'}(X))$ by Bayes' rule (Appendix 13.2). Three corrections appear: the outcome residual reweighted to the mediator distribution of arm d' , the nested residual reweighted by the arm d' propensity, and the plug-in.

7.2 Orthogonality and rates

Proposition 2. $E[\psi_{d,d'}] = E[Y(d, M(d'))]$ under sequential ignorability, and the Gateaux derivative of $E[\psi_{d,d'}]$ with respect to each of the nuisances $(\mu_d, p_d(X), p_d(M, X), \nu_{d,d'})$ at their true values is zero. The estimator $\bar{\psi}_{d,d'}$ with cross-fitted nuisances is root- n consistent and asymptotically normal when the products of the nuisance errors, $\|\hat{\mu}_d - \mu_d\| \cdot \|\hat{p} - p\|$ and $\|\hat{\nu} - \nu\| \cdot \|\hat{p} - p\|$, are $o(n^{-1/2})$.

Skeleton. Take expectations term by term. The first term has mean zero because $E[Y - \mu_d(M, X) \mid D = d, M, X] = 0$; the second because $E[\mu_d(M, X) - \nu_{d,d'}(X) \mid D = d', X] = 0$; the third is the mediation formula. Perturb one nuisance at a time: each first-order term is killed by one of the two conditional-mean-zero facts, and what remains is a product of two errors. Cross-fitting removes the dependence between the nuisance fits and the scores. Appendix 13.2 gives the algebra.

The estimator is triply robust in the sense of Tchetgen Tchetgen and Shpitser: it is consistent if any two of the three nuisance groups (outcome model, mediator-related propensities, treatment propensity) are correct.

7.3 Cross-fitting and a Monte Carlo

`mediate_dml()` cross-fits all nuisances with `mlr3` learners. The nested regression is the delicate step. In each fold the outcome model is trained on the $D = d$ units of the training folds; its predictions on the $D = d'$ units of the same training folds become the labels of the nested regression; and that regression is evaluated on the held-out fold. No observation's label enters its own nuisance. The default learners are linear and logistic regressions, which reproduce the linear system.

```
fit_dml <- mediate_dml(dat, "y", "d", "m", x = c("x1", "x2"), seed = 1)
fit_dml$effects[1:3, ]
#>   term estimate std.error conf.low conf.high
#> 1 total 2.070118 0.04976798 1.9725743 2.167661
#> 2 nde 1.038788 0.05763044 0.9258348 1.151742
#> 3 nie 1.031329 0.05592683 0.9217148 1.140944
```

The `nonlinear` design has $\exp(0.5x_1)$, x_2^2 , $\cos(x_1)$, and x_1x_2 in the mediator and outcome models, and a treatment-mediator interaction. Table 4 reports 50 simulations of `mediate_reg()` with an interaction, `mediate_dml()` with linear learners, and `mediate_dml()` with random forests.

Table 4: Nonlinear design, $n = 1500$, 50 simulations: regression with interaction (simulation intervals), double machine learning with linear learners, and with random forests. Coverage of 95 percent intervals.

Effect	Estimator	Truth	Bias	SD	RMSE	Coverage
NDE	Regression	1.334	0.102	0.124	0.159	0.88
	DML linear	1.334	0.091	0.232	0.247	0.92
	DML forest	1.334	0.029	0.188	0.188	0.90
NIE	Regression	1.560	0.527	0.137	0.544	0.00
	DML linear	1.560	0.393	0.227	0.453	0.50
	DML forest	1.560	-0.012	0.185	0.183	0.92

The regression has an NIE bias of 0.53 and coverage 0%: a wrong mediator model puts the wrong distribution into the formula. Linear learners inside the influence function reduce the NIE bias to 0.39, because the propensity corrections repair part of the outcome model's error, but the corrections are themselves misspecified, so the bias does not vanish and the NDE bias (0.09) is about the regression's (0.10). Orthogonality protects against small nuisance errors, not against wrong models. Forests bring the biases to 0.03 and -0.01 with coverage 90% and 92%. Figure 7 shows the sampling distributions.

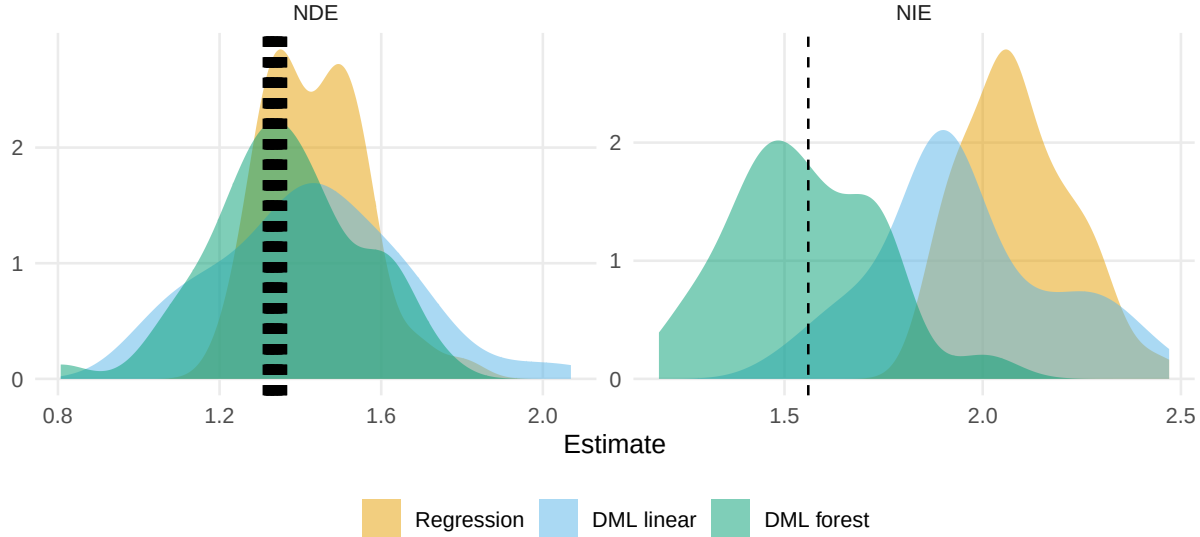


Figure 7: Sampling distributions of the natural direct and indirect effects in the nonlinear design. Dashed lines are the truth.

7.4 Controlled direct effects and treatment-induced confounders

With a binary mediator, $E[Y(d, m)]$ has an AIPW score with the joint propensity $P(D = d, M = m | X)$, and `mediate_dml(m_ref = )` adds the controlled direct effect at m .

```
fit_dml_bin <- mediate_dml(dat_bin, "y", "d", "m", x = c("x1", "x2"), m_ref = 0, seed = 1)
fit_dml_bin$effects[fit_dml_bin$effects$term %in% c("nde", "cde"), ]
#>   term estimate std.error   conf.low conf.high
#> 2  nde 0.1650963 0.01996781 0.12596013 0.2042325
#> 6  cde 0.1535798 0.02949163 0.09577727 0.2113823
attr(dat_bin, "truth")$cde
#> [1] 0.165
```

When a treatment-induced confounder L is present, Acharya, Blackwell, and Sen (2016) estimate the controlled direct effect in two steps. Regress Y on D , M , X , and L ; the mediator coefficient $\hat{\delta}$ is identified because L is in the regression. Then form the demediated outcome $\tilde{Y} = Y - \hat{\delta}(M - m)$ and regress it on D and X only; L is out, so the coefficient on D is not a post-treatment partial coefficient. The bootstrap of both steps gives the standard errors. The `post_treatment_confounder` design has such an L (called z in the data).

```
dat_z <- sim_mediation(3000, dgp = "post_treatment_confounder", seed = 7)
fit_cde <- mediate_cde(dat_z, "y", "d", "m", x_pre = c("x1", "x2"), x_post = "z", n_boot = 199, seed = 1)
fit_cde$effects[fit_cde$effects$term %in% c("cde", "naive_direct"), ]
#>   term estimate std.error   conf.low conf.high
#> 1      cde 1.1415844 0.05758089 1.0222086 1.2436281
#> 2 naive_direct 0.6378025 0.04669567 0.5427547 0.7233829
attr(dat_z, "truth")$cde
#> [1] 1.09
```

The single regression that conditions on L and M together gives 0.64 for a truth of 1.09: it removes the part of the direct effect that runs through L . The natural effects are not identified in this design at all, which is what the `truth` attribute records.

8 Mechanism accounting in regressions

8.1 Add the mediators and watch the coefficient

The most common mechanism exercise in applied economics adds candidate mediators to the outcome regression and reports how far the treatment coefficient falls. Figure 8 does this in a design with two mediators that carry indirect effects of 0.8 and 0.6, a direct effect of 0.5, and mediator errors that share a common component, as they do whenever two channels respond to the same unobserved trait. The problem is visible: the drop attributed to each mediator depends on the order of addition, because the second mediator to enter only gets the part of the change that the first did not already absorb.

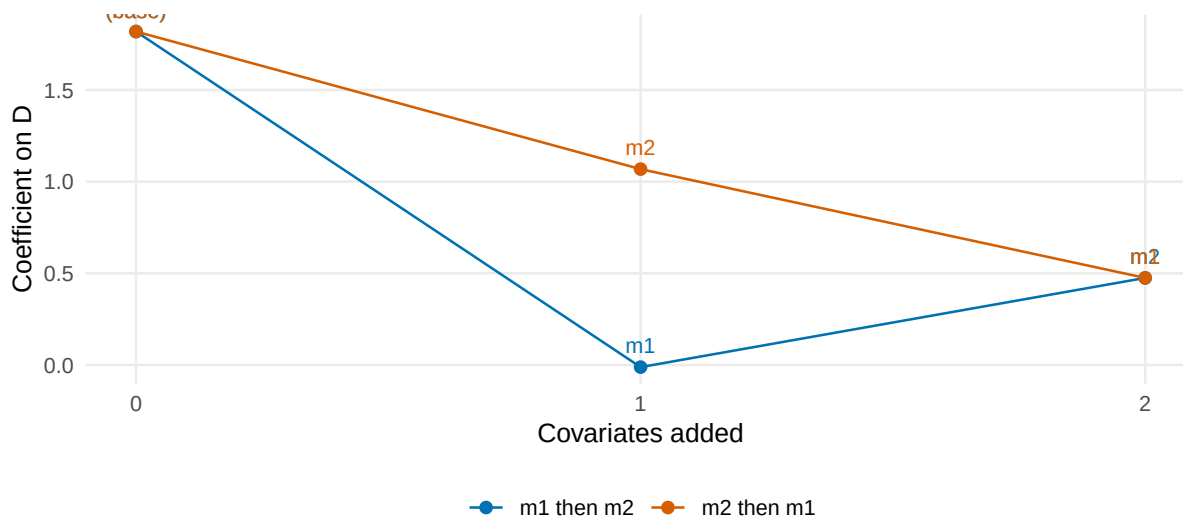


Figure 8: The treatment coefficient as mediators are added one at a time, in two orders, in the correlated-mediators design ($n = 3000$). The drop attributed to $m1$ is different when it enters first and when it enters second.

Adding $m1$ first drops the coefficient by 1.83; adding it second drops it by 0.59. The indirect effect through $m1$ is 0.8, and neither number has a claim to it.

8.2 Gelbach's decomposition

Gelbach (2016) shows that the omitted-variable-bias formula assigns the change to each added covariate in a unique, order-independent way. With the base regression $Y \sim D + X_{base}$ and the full regression that adds $X_{add} = (Z_1, \dots, Z_K)$,

$$\hat{\beta}^{base} - \hat{\beta}^{full} = \sum_{k=1}^K \hat{\Gamma}_k \hat{\beta}_k^{full},$$

where $\hat{\Gamma}_k$ is the coefficient on D in the auxiliary regression $Z_k \sim D + X_{base}$ and $\hat{\beta}_k^{full}$ is the coefficient of Z_k in the full regression. The identity is exact in the sample (Appendix 13.4). Each term is the part of the change that covariate k accounts for. In the design of Figure 8 each term is also the indirect effect through that mediator, because the outcome model is linear and both mediators are in the full regression, so their coefficients are identified despite the shared error.

```
g_seq <- gelbach_decomp(dat_seq, "y", "d", x_base = c("x1", "x2"), x_add = c("m1", "m2"), n_boot = 199, seed = 1)
g_seq
#> Gelbach decomposition of the coefficient on d (n = 3000)
#>   term estimate std.error conf.low conf.high
#>   base   1.8186   0.10052   1.6255   2.0106
#>   full   0.4746   0.03935   0.3927   0.5501
#>   change  1.3439   0.09614   1.1578   1.5378
```

```
#> Contributions of the added covariates (they sum to the change; identity gap -2.2e-15):
#> covariate Gamma beta_full contribution std.error conf.low conf.high share
#>      m1 0.9272  0.8118    0.7528  0.03584  0.6836  0.8277 0.5601
#>      m2 0.3973  1.4881    0.5912  0.07212  0.4580  0.7306 0.4399
```

The decomposition is bookkeeping. It says how a coefficient moves when covariates enter; it says nothing about why unless the full regression identifies the mediator effects. That is the same condition as before.

8.3 Wheeler's mechanisms revisited

Table 6 of Wheeler et al. (2022) reports treatment effects on candidate mechanisms. Of the public ones, four are individual outcomes (placed in a targeted sector, placed through a partner firm, self-placed, completed the program) and two are indices (engagement, aspirations). A Gelbach decomposition of the end-of-program employment effect on these six variables tells us which of them absorb the coefficient. The indices are post-treatment; the decomposition is an accounting, in the sense of Section 2.3.

```
mechs <- c("li_account", "targetsector", "harambee_partner_original", "selfplaced", "s2_completed", "summary_engage")
wh_g <- wh[complete.cases(wh[, c("employed1_all", "treatment", mechs)]), ]
g_wh <- gelbach_decomp(wh_g, "employed1_all", "treatment", x_add = mechs, fe = "strata",
                      cluster = "studref", n_boot = 199, seed = 1)

g_wh$coefficients
#>      term estimate std.error conf.low conf.high
#> 1 base 0.08712918 0.04232274 0.01648579 0.166145703
#> 2 full -0.04155684 0.03306185 -0.12334007 0.002130525
#> 3 change 0.12868601 0.04321429 0.05510206 0.206916724
g_wh$contributions[, c("covariate", "Gamma", "beta_full", "contribution", "share")]
#>      covariate Gamma beta_full contribution share
#> 4 li_account 0.333229767 0.02775508 0.009248818 0.07187120
#> 5 targetsector 0.191189859 0.62182588 0.118886802 0.92385177
#> 6 harambee_partner_original -0.078463419 0.03565020 -0.002797237 -0.02173691
#> 7 selfplaced -0.008490273 0.55515789 -0.004713442 -0.03662746
#> 8 s2_completed -0.019043612 -0.22067649 0.004202478 0.03265683
#> 9 summary_engage 0.152616509 0.02528295 0.003858596 0.02998458
plot_gelbach(g_wh)
```

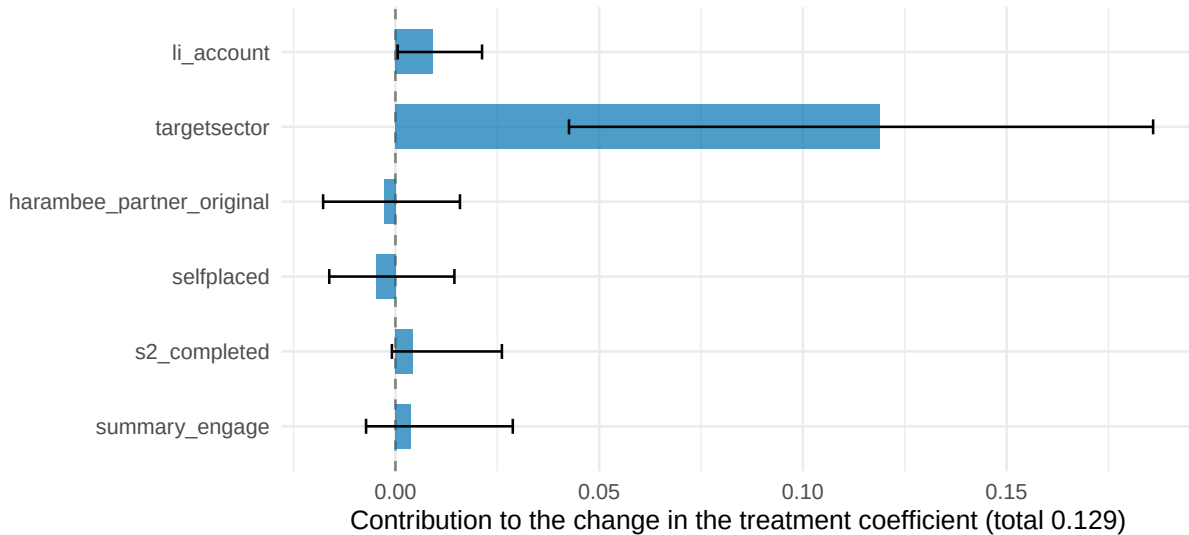


Figure 9: Gelbach decomposition of the treatment coefficient on end-of-program employment in Wheeler et al. (2022), with the LinkedIn account indicator and five public mechanism variables added to the block-fixed-effects regression. Bars are contributions with cluster-bootstrap intervals.

The base coefficient of 0.087 falls to -0.042 once the six variables enter. The largest single contribution is from **targetsector** (0.119, or 92% of the change). Placement channels are outcomes of the same job search that produces employment, so their

contributions describe where the jobs came from rather than a mechanism that could be manipulated. The LinkedIn account now contributes 0.009, or 7% of the change, against the 0.73 share of Section 5.2: with the placement variables in the regression, the account's own coefficient is small because the placement variables sit downstream of it and absorb its effect. The order-invariant decomposition does not remove that dependence on which covariates are included.

9 Endogenous mediators

9.1 The instrumental-variables system

When the mediator is endogenous but an instrument Z shifts it, Cattán et al. (2025) estimate three regressions with the same controls:

$$\begin{aligned} M &= \delta_0 + \delta_1 D + \delta_2 Z + X' \delta_X + e_M && \text{(first stage),} \\ Y &= \gamma_0 + \gamma_1 D + \gamma_2 M + X' \gamma_X + e_Y, \quad M \text{ instrumented by } Z && \text{(second stage),} \\ Y &= \pi_0 + \pi_1 D + X' \pi_X + e && \text{(reduced form).} \end{aligned}$$

The direct effect is γ_1 , the indirect effect is $\delta_1 \gamma_2$, and the total effect is π_1 . Adding Z to the reduced form, the coefficient on D equals $\gamma_1 + \delta_1 \gamma_2$ exactly (Appendix 13.3). Beyond instrument validity given D and X , the decomposition needs the mediator effect γ_2 to be the same for the instrument's compliers and for the units whose mediator D moves. Huber (2019) and Dippel, Ferrara, and Heblich (2020) call this homogeneity; Cattán et al. check it by re-estimating γ_2 in subgroups (their Figure A13).

The `iv_mediator` design has an unobserved U that raises both M and Y , a binary instrument, direct effect 0.8, and indirect effect $-0.6 \times 0.7 = -0.42$: a positive total made of opposite-signed parts, as in the elite-peers paper.

```
dat_iv <- sim_mediation(3000, dgp = "iv_mediator", seed = 6)
fit_iv <- mediate_iv(dat_iv, "y", "d", "m", z = "z", x = c("x1", "x2"), n_boot = 199, seed = 1)
fit_iv
#> Mediation with an instrumented mediator (m instrumented by z; n = 3000)
#>               term estimate std.error conf.low conf.high
#>               total   0.3340   0.07809   0.1756   0.4738
#>               total_z  0.3352   0.07761   0.1854   0.4856
#>               direct   0.7480   0.06082   0.6323   0.8751
#>               indirect -0.4128   0.04705  -0.5224  -0.3367
#>               first_stage -0.6315   0.05193  -0.7368  -0.5468
#>               mediator_effect  0.6537   0.05315   0.5438   0.7421
#>               naive_direct  1.0508   0.04541   0.9656   1.1416
#>               naive_mediator_effect  1.1318   0.01534   1.1026   1.1620
#>               first-stage F for the instrument(s): 338.1
```

The naive mediator coefficient is 1.13 for a truth of 0.7, and the naive direct effect 1.05 for a truth of 0.8: U inflates the mediator's coefficient and the collider at M moves the treatment's. The instrumented system gives 0.65 (standard error 0.05) and 0.75 (standard error 0.06), which are 0.9 and 0.9 standard errors from the truth. Across 30 further samples of the design the direct effect averages 0.81 with standard deviation 0.07, so the estimator is centred and the bootstrap standard error is about right. Figure 10 shows the decomposition, and the homogeneity check below re-estimates the mediator effect by subgroup.

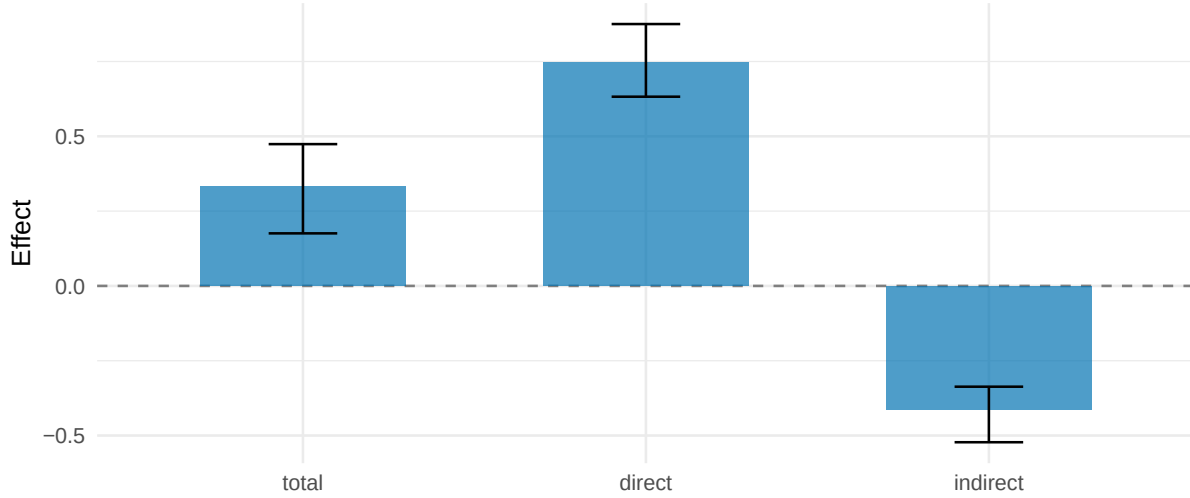


Figure 10: Total, direct, and indirect effects in the instrumented-mediator design, with bootstrap intervals. The total effect is small because the two channels have opposite signs.

```
dat_iv$group <- ifelse(dat_iv$x2 > 0, "x2 > 0", "x2 <= 0")
mediate_iv(dat_iv, "y", "d", "m", z = "z", x = c("x1", "x2"), homogeneity_by = "group", n_boot = 49)$homogeneity
#>      group      n mediator_effect direct first_stage
#> 1 x2 <= 0 1496      0.6892607 0.6853002 -0.6819713
#> 2 x2 > 0 1504      0.6075145 0.8114105 -0.5758505
```

9.2 Front-door adjustment

The right panel of Figure 3 has no valid back-door adjustment set: U is unobserved. Pearl (1995) shows that the effect of D on Y is still identified when a mediator satisfies three conditions: M intercepts every directed path from D to Y ; no unblocked back-door path from D to M ; and every back-door path from M to Y is blocked by D . Then

$$E[Y(d)] = \sum_m P(m | d) \sum_{d'} E[Y | d', m] P(d'),$$

with covariates X added to every conditioning set (Appendix 13.5). The formula chains two identified effects: D on M (no confounding), and M on Y with D as the adjustment variable (because D blocks $M \leftarrow D \leftarrow U \rightarrow Y$).

Bellemare, Bloem, and Wexler (2020) give the applied reading with a ride-sharing example: the effect of a driver's tipping prompt on tips, with the mediator being whether the rider saw the prompt. The mediator must carry the whole effect, and that is the assumption to argue. `front_door()` estimates the formula by two regressions with a bootstrap, or by the influence-function estimator of Fulcher et al. (2020) for a binary mediator.

```
dat_fd <- sim_mediation(4000, dgp = "front_door", seed = 9)
fd_reg <- front_door(dat_fd, "y", "d", "m", x = c("x1", "x2"), n_boot = 199, seed = 1)
fd_reg$effects[1:2, ]
#>      term estimate std.error conf.low conf.high
#> 1 total 0.5881448 0.03097177 0.5335522 0.6524895
#> 2 backdoor 1.6082944 0.05738277 1.4976822 1.7171943
fd_aipw <- front_door(dat_fd, "y", "d", "m", x = c("x1", "x2"), method = "aipw", seed = 1)
fd_aipw$effects[1, ]
#>      term estimate std.error conf.low conf.high
#> 1 total 0.5878941 0.03108695 0.5269648 0.6488234
attr(dat_fd, "truth")$total
#> [1] 0.5445
```

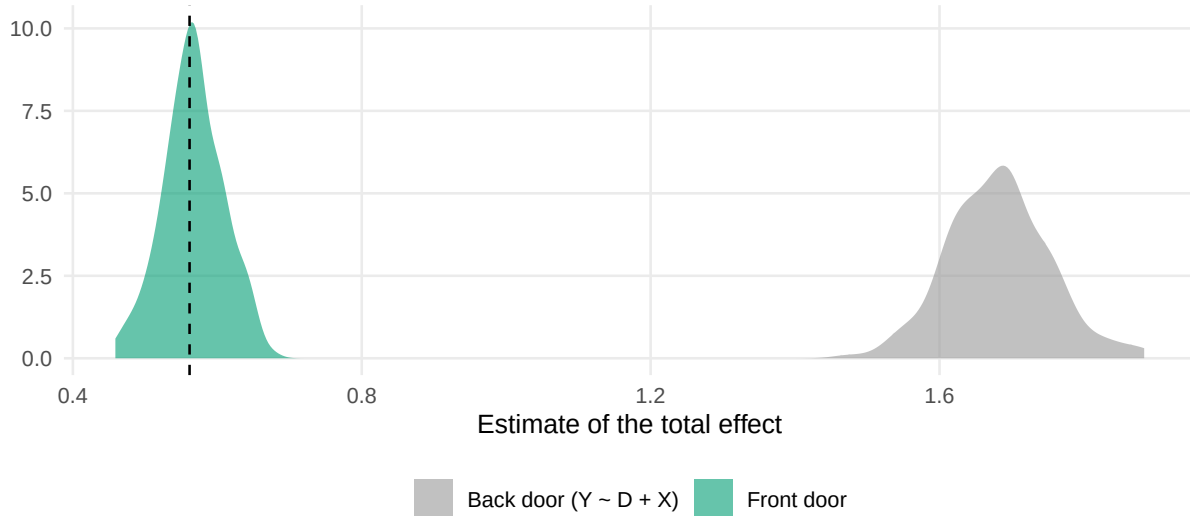


Figure 11: Back-door regression and front-door estimate across repeated samples of the front-door design ($n = 2000$). The back-door regression adjusts for X only and is biased by the unobserved U ; the front-door estimate centres on the truth (dashed).

9.3 What the two have in common

The instrument and the front door both give the mediator a source of variation that the outcome equation cannot see except through the mediator. With an instrument, that source is Z . With the front door, it is the treatment itself, once we grant that D has no other route to Y . In both cases the estimator multiplies an effect of something on M by an effect of M on Y estimated with a different conditioning set. The credibility of the result is the credibility of the exclusion.

10 Multiple and latent mediators

10.1 Parallel mediators

With mediators $M_1, \dots, M_J$ that do not cause each other, the joint natural indirect effect is $E[Y(1, M_1(1), \dots, M_J(1)) - Y(1, M_1(0), \dots, M_J(0))]$, and the mediator-specific effect moves one mediator at a time. `mediate_reg()` accepts a vector of mediators and reports both. The sum of the specific effects equals the joint effect when the outcome model is additive in the mediators.

```
dat_par <- sim_mediation(3000, dgp = "parallel", seed = 4)
fit_par <- mediate_reg(dat_par, "y", "d", c("m1", "m2"), x = c("x1", "x2"), n_sim = 500, seed = 1)
fit_par$effects[fit_par$effects$term %in% c("total", "nde", "nie", "nie_m1", "nie_m2"), ]
#>   term estimate std.error conf.low conf.high
#> 1 total 1.9059405 0.07398412 1.7564669 2.0479020
#> 2 nde 0.4841785 0.04080620 0.4043329 0.5648065
#> 3 nie 1.4217620 0.07096127 1.2845164 1.5537156
#> 7 nie_m1 0.7439141 0.03636445 0.6738650 0.8188299
#> 8 nie_m2 0.6778478 0.05771048 0.5653926 0.7932462
unlist(attr(dat_par, "truth")[c("nde", "nie", "nie_m1", "nie_m2")])
#>   nde   nie nie_m1 nie_m2
#> 0.5  1.4  0.8  0.6
```

Sequential ignorability must now hold for the mediator vector, and it must be that no mediator affects another. If M_1 causes M_2 , the effect through M_1 includes the part relayed by M_2 , and the mediator-specific effects need the path-specific machinery of Avin, Shpitser, and Pearl (2005), which is beyond this section. Figure 12 draws the assumption.

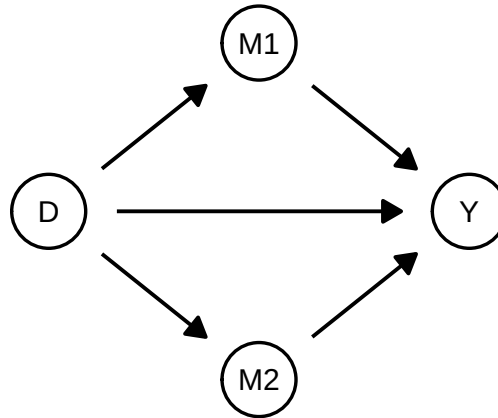


Figure 12: Parallel mediators: each is caused by D and causes Y, and neither causes the other.

10.2 Mediators from text

Zhang, Li, and Allenby (2024) ask why consumers reject cultivated meat. The treatments are information conditions, the outcome is a purchase choice, and the mediators are what respondents wrote in an open-ended answer. A seeded topic model turns each answer into shares of three topics fixed in advance (animal welfare, disgust, sustainability) and one free topic. The treatment shifts the prior over topic shares, and the shares enter a logistic outcome model, with an optional direct path. Everything is estimated in one Markov chain Monte Carlo run.

The structure is the parallel-mediator model of the previous subsection with mediators that are latent. Two ways to estimate it:

- **Integrated.** The authors' approach. The topic shares and the mediation parameters are drawn jointly, so the uncertainty in the mediator construction propagates into the effects.
- **Two-step.** Fit the seeded topic model, take posterior mean shares as mediator values, and run `mediate_reg()` with a logit outcome and the shares as parallel mediators. Simpler and transparent, but the shares are measured with error, and measurement error in a mediator attenuates its coefficient and pushes the missing part into the direct effect.

The replication note in the notes repository runs the two-step version on their data and compares it with the paper's Tables 4 to 7. For the lecture the point is the identification burden, which is the same as for any mediator: the topics must be unconfounded with the choice given the treatment. Text is measured after treatment and describes reasons the respondent chose to give, so the assumption is that what they wrote and what they chose share no unmeasured cause beyond the treatment.

11 Sensitivity, design, and credibility

11.1 Reporting sensitivity

A mediation estimate under sequential ignorability should come with the value of ρ_0 from Section 6.3, or with an equivalent bound. VanderWeele (2015, ch. 3) gives bias formulas in terms of the strength of an unmeasured U on Y and its prevalence difference across mediator levels. Ding and VanderWeele's E-value has a mediation version: the minimum strength of association that U needs with both M and Y to explain away the indirect effect. All three say the same thing in different units. The lecture's advice is to report one and to compare it with the strength of the observed covariates, which puts the number on a scale the reader knows.

11.2 Designing for mechanisms

- Measure candidate mediators before the outcome, and pre-register the list. Wheeler et al. note that none of their mechanism analysis was prespecified; a registered list turns a mechanism story into a test.
- Randomize an encouragement for the mediator whenever the budget allows a second arm. An instrument for M is worth more than any covariate.
- When the mediator is a choice, look for institutional randomness that moves it, as the exam lottery does for grades.
- Collect the intermediate confounders. If a treatment-induced L is plausible, the controlled direct effect is the estimand to report.

11.3 Graphs as statements, not as evidence

Imbens (2020) argues that graphs have had little influence in economics because the identifying restrictions economists trust are about behaviour and design, not about arrows between variables. The reply from the graphical side is that a graph makes those restrictions explicit and checkable. Both are right. In this section the graph did two jobs: it showed why a post-treatment control fails, and it listed the assumptions that an estimator needs. It cannot tell us whether U exists. The credibility of a mediation claim rests on the argument for SI-2, the instrument, or the exclusion, and the graph is the place to write that argument down.

12 Practice

12.1 Replications and anchors

Three papers anchor this section; their replication notes live next to the lecture in the notes repository.

- **Wheeler, Garlick, Johnson, Shaw, and Gargano (2022), LinkedIn training.** The public data reproduce Table C1 with `mediate_reg()` (Section 5.2) and support the Gelbach decomposition of Section 8.3. The LinkedIn usage measures that build the intensive-margin indices are not in the public release.
- **Cattan, Salvanes, and Tominey (2025), elite peers.** The registry data are not public. The replication note audits the code path and runs `mediate_iv()` on the authors' simulated data to reproduce the structure of their Table 4: first stage, instrumented second stage, and the decomposition panel.
- **Zhang, Li, and Allenby (2024), text mediators.** The data and seed lists are public. The replication note fits the seeded topic model and runs the two-step version with `mediate_reg()`, next to the authors' integrated estimates.

12.2 A mechanisms checklist

1. State which estimand answers the question: total, controlled direct, natural direct and indirect, or an accounting of a coefficient change.
2. Draw the graph, mark what is observed, and say where sequential ignorability could fail. Name the candidate U and L .
3. If D is randomized, report the total effect and the effect on each candidate mediator first. These need no mediation assumption.
4. For natural effects, fit `mediate_reg()` with and without the interaction and `mediate_dml()` with flexible learners; report both decompositions and the proportion mediated with its standard error.
5. Report the sensitivity parameter ρ_0 and compare it with the observed covariates' strength.
6. If an instrument for the mediator exists, run `mediate_iv()` and the homogeneity check. If the front-door structure is credible, run `front_door()` and argue the exclusion.
7. If a treatment-induced confounder exists, report the controlled direct effect from `mediate_cde()` and say that natural effects are not identified.
8. For regression decompositions, use `gelbach_decomp()` rather than sequential addition, and label the result as accounting unless the full regression is argued to identify the mediator effects.

13 Appendix

13.1 The mediation formula under sequential ignorability

Fix x . By SI-1, $M(d') \perp D \mid X$, so the distribution of $M(d')$ given $X = x$ equals the distribution of $M(d')$ given $D = d', X = x$, which by consistency is the distribution of M given $D = d', X = x$. Next,

$$E[Y(d, M(d')) \mid X = x] = \int E[Y(d, m) \mid M(d') = m, X = x] dF_{M(d') \mid X=x}(m).$$

By SI-1 again, $\{Y(d, m), M(d')\} \perp D \mid X$, so conditioning on $D = d$ changes nothing: $E[Y(d, m) \mid M(d') = m, X = x] = E[Y(d, m) \mid M(d') = m, D = d, X = x]$. By SI-2, $Y(d, m) \perp M(d') \mid D = d, X = x$, so this equals $E[Y(d, m) \mid D = d, X = x]$. A third application of SI-2 with $d' = d$ gives $Y(d, m) \perp M(d) \mid D = d, X = x$, hence $E[Y(d, m) \mid D = d, X = x] = E[Y(d, m) \mid M(d) = m, D = d, X = x] = E[Y \mid D = d, M = m, X = x]$ by consistency. Substitute and integrate over x .

The positivity conditions ensure that every conditional expectation is defined on the support of (D, M, X) .

13.2 The efficient influence function and its orthogonality

Bayes' rule for the density ratio. $f(M \mid d', X)/f(M \mid d, X) = \{P(D = d' \mid M, X)/P(D = d')\} \cdot \{P(D = d \mid X)/P(D = d \mid M, X)\}$ after cancelling $f(M \mid X)$, so

$$\frac{f(M \mid d', X)}{f(M \mid d, X) p_{d'}(X)} = \frac{p_{d'}(M, X)}{p_d(M, X) p_{d'}(X)} \cdot \frac{p_d(X)}{p_{d'}(X)} \cdot \frac{p_{d'}(X)}{p_d(X)},$$

and the last two factors cancel. This is the weight in $\psi_{d,d'}$.

Unbiasedness. Write $\psi_{d,d'} = A + B + \nu_{d,d'}(X)$ with A the outcome-residual term and B the nested-residual term. $E[A \mid D = d, M, X] = w(M, X) E[Y - \mu_d(M, X) \mid D = d, M, X] = 0$ and $A = 0$ when $D \neq d$, so $E[A] = 0$. $E[B \mid D = d', X] = p_{d'}(X)^{-1} E[\mu_d(M, X) - \nu_{d,d'}(X) \mid D = d', X] = 0$ by the definition of ν , so $E[B] = 0$. $E[\nu_{d,d'}(X)] = E[Y(d, M(d'))]$ by the mediation formula.

Orthogonality. Perturb the nuisances by ϵ in the direction of a fixed alternative and differentiate $E[\psi_{d,d'}]$ at $\epsilon = 0$.

- Perturbing μ_d by ϵh : the change in A is $-\epsilon E[w(M, X) 1\{D = d\} h(M, X)] = -\epsilon E[p_d(X) w h]$ after conditioning on (M, X) ; the change in B is $+\epsilon E[1\{D = d'\} p_{d'}(X)^{-1} h(M, X)] = \epsilon E[E[h(M, X) \mid D = d', X]]$. Using the density-ratio identity, $E[p_d(X) w(M, X) h(M, X) \mid X] = E[h(M, X) \mid D = d', X]$, so the two changes cancel.
- Perturbing $\nu_{d,d'}$ by ϵk : B changes by $-\epsilon E[1\{D = d'\} p_{d'}(X)^{-1} k(X)] = -\epsilon E[k(X)]$ and the plug-in by $+\epsilon E[k(X)]$.
- Perturbing the propensities: the weights multiply residuals whose conditional means are zero at the true μ and ν , so the derivative is zero.

The remainder after the first-order terms is a sum of products of two nuisance errors, which gives the rate condition of Proposition 2. The cross-fitted estimator inherits the asymptotic variance $\text{var}(\psi_{d,d'})$, and the effects are differences of the four $\bar{\psi}_{d,d'}$ with variances computed from the differences of the influence functions.

Controlled direct effect. For a binary mediator, $E[Y(d, m)]$ is identified by $E[E[Y \mid D = d, M = m, X]]$ under the intervention-world assumptions, and its influence function is the AIPW score with the joint propensity $P(D = d, M = m \mid X) = p_d(X) P(M = m \mid D = d, X)$.

13.3 Delta-method variances for products and ratios

For an estimator $\hat{\theta}$ of $\theta = (\theta_1, \dots, \theta_K)$ with covariance V and a differentiable g , $\widehat{\text{var}}(g(\hat{\theta})) = \nabla g(\hat{\theta})' V \nabla g(\hat{\theta})$.

- *Sobel.* $g = \alpha_1 \beta_2$ with α_1 and β_2 from separate regressions treated as independent: $\nabla g = (\beta_2, \alpha_1)$, so $\widehat{\text{var}} = \beta_2^2 v_{\alpha_1} + \alpha_1^2 v_{\beta_2}$. The two regressions share the sample, but their coefficient estimates are

asymptotically uncorrelated when the mediator model is correctly specified, which is why the formula works in the linear case.

- *Shares.* $S_1 = \alpha\gamma/\beta$ with (β, γ, α) from three regressions on the same sample: $\nabla S_1 = S_1(-1/\beta, 1/\gamma, 1/\alpha)$. The covariance V across regressions is the covariance of the stacked influence functions $n^{-1} \sum_i (X_i' X_i / n)^{-1} X_i e_i$ of the three coefficient vectors, clustered when needed. `mediate_reg()` computes it this way; it is what Stata's `suest` does.
- *IV decomposition.* With the reduced form including Z , $Y = \pi_0 + \pi_1 D + \pi_2 Z + X' \pi_X + e$, two-stage least squares satisfies $\hat{\pi}_1 = \hat{\gamma}_1 + \hat{\gamma}_2 \hat{\delta}_1$ and $\hat{\pi}_2 = \hat{\gamma}_2 \hat{\delta}_2$ exactly, because the second-stage fitted values are linear in the first-stage coefficients. The bootstrap of the whole system in `mediate_iv()` gives the variance of $\hat{\delta}_1 \hat{\gamma}_2$ without the cross-equation covariance formula.

13.4 Gelbach's identity

Let $W = (1, D, X_{base})$ and Z the $n \times K$ matrix of added covariates. The full regression $Y = Wb + Zc + e$ with OLS residual e orthogonal to W and Z . Regress both sides on W alone: the OLS coefficient of Y on W is $b + (W'W)^{-1}W'Zc + (W'W)^{-1}W'e$, and the last term is zero. The middle term is $\sum_k \hat{\Gamma}_k c_k$ where $\hat{\Gamma}_k = (W'W)^{-1}W'Z_k$ is the coefficient vector of Z_k on W . Taking the D component gives $\hat{\beta}^{base} - \hat{\beta}^{full} = \sum_k \hat{\Gamma}_{k,D} \hat{c}_k$. The identity is algebraic, so it holds in every sample and for every ordering of the added covariates. With fixed effects, replace all variables by their within-transformed versions; `gelbach_decomp()` does this through `fixest`.

13.5 Front-door identification

In the graph $D \rightarrow M \rightarrow Y$, $D \leftarrow U \rightarrow Y$, with U unobserved. The effect of D on M is identified by $P(m | d)$, because the only back-door path $D \leftarrow U \rightarrow Y \leftarrow M$ is blocked at the collider Y . The effect of M on Y is identified by adjusting for D , because D blocks the back-door path $M \leftarrow D \leftarrow U \rightarrow Y$ and D is not a descendant of M : $P(y | do(m)) = \sum_{d'} P(y | m, d') P(d')$. Chaining, since D affects Y only through M , $P(y | do(d)) = \sum_m P(m | d) P(y | do(m)) = \sum_m P(m | d) \sum_{d'} P(y | m, d') P(d')$. Taking expectations gives the formula of Section 9.2; adding observed covariates X to every term is immediate. The influence function of Fulcher et al. (2020) for $E[Y(d)]$ with a binary mediator is

$$\varphi_d = \frac{f(M | d, X)}{f(M | D, X)} (Y - \mu(D, M, X)) + \frac{1\{D = d\}}{p_d(X)} (\xi(M, X) - \eta_d(X)) + \eta_d(X),$$

with $\mu(d', m, x) = E[Y | d', m, x]$, $\xi(m, x) = \sum_{d'} \mu(d', m, x) P(d' | x)$, and $\eta_d(x) = \sum_m f(m | d, x) \xi(m, x)$; its orthogonality follows the same pattern as Appendix 13.2.

13.6 References

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